1/8

OSN

OS : Coordinates everything

↳ Middleman b/w Hardware and user

OS handles interactions with the disk and performs storage management.

2/8

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main(){
    printf("This is parent. PID: %d\n",(int)getpid());
    int rc=fork();
    if(rc<0){
        printf("Fork failed!\n");
        exit(1);
    }
    else if(rc==0){
        printf("This is child. PID: %d\n",(int)getpid());
    }
    else{
        printf("This is parent my child is %d\n",rc);
    }
    printf("Who am I? PID:%d\n",(int)getpid());
    return 0;
}
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    printf("This is parent. PID: %d\n",(int)getpid());
    int rc=fork();
    if(rc<0){
        printf("Fork failed!\n");
        exit(1);
    }
    else if(rc==0){
        printf("This is child. PID: %d\n",(int)getpid());
    }
    else{
        int rc_wait=wait(NULL);
        printf("This is parent my child is %d\n",rc);
    }
    printf("Who am I? PID:%d\n",(int)getpid());
    return 0;
}
```

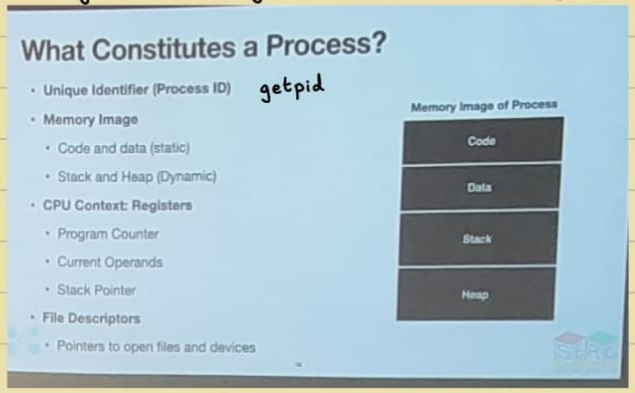
```c
    }
    printf("Who am I? PID:%d\n",(int)getpid());
    return 0;
}

// wait() makes the parent wait for any one of the child and returns the pid of chi
// we can use wait() in a loop to wait for all children (loop will it doesnt return
// waitpid(child_pid) waits for the specified child
// wait(), waitpid() only works for pareth-child pair where parent waits for the ch
```

5/8

OS → divides CPU work, makes it feel like any process has infinite access to CPU.

Process → Takes resources

- Memory :
  - Static Memory → Variables [Stack]
  - Dynamic Memory → malloc, calloc [Heap]

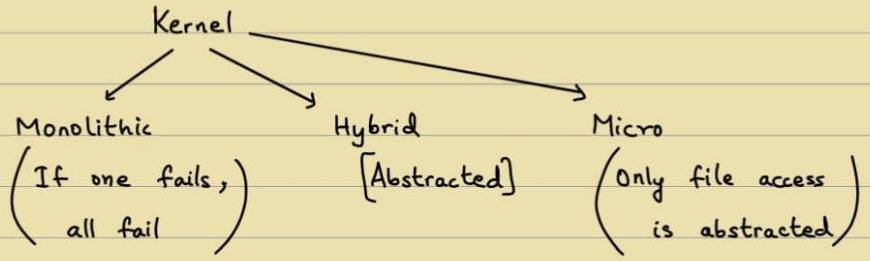


Create memory image for the process

⇓

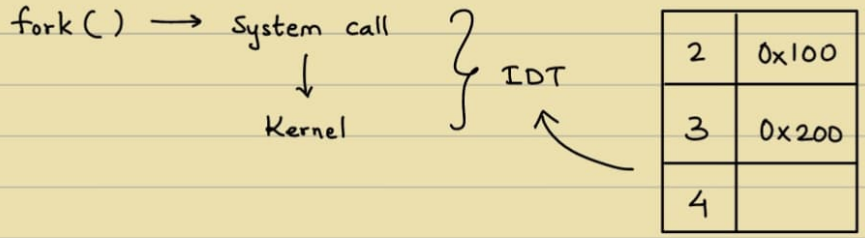
Allocate PC, SP

- Kernel : Handles direct instructions with hardware
  - ↳ Part of the code



Kernel has stack
↳ Process

- API → System calls (fork, exec, wait)
  - ↳ Resides in Kernel
    - Limited Direct Execution (LDE) → CPU



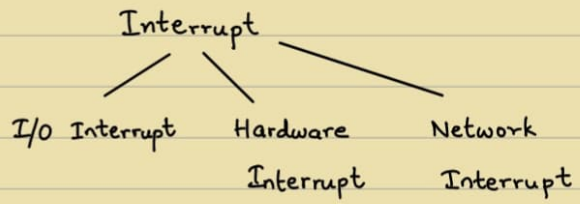
System call ⇒ Mode change ⇒ User mode to Kernel Mode
↳ TRAP Instruction : Changes privilege

IDT ⇒ OS code gets executed
↳ Interrupt disrupt Table

Context → PC, SP → Saved

User stack ✗
Kernel stack ✓
↳ Per process level

```
10
11  syscall  <- OS
12
```



- Context Switch:

  How to decide which process to run next ⟶ Scheduling

  In reality, Jobs can arrive at any time.

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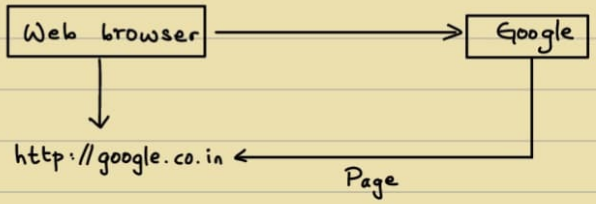


Port → 80

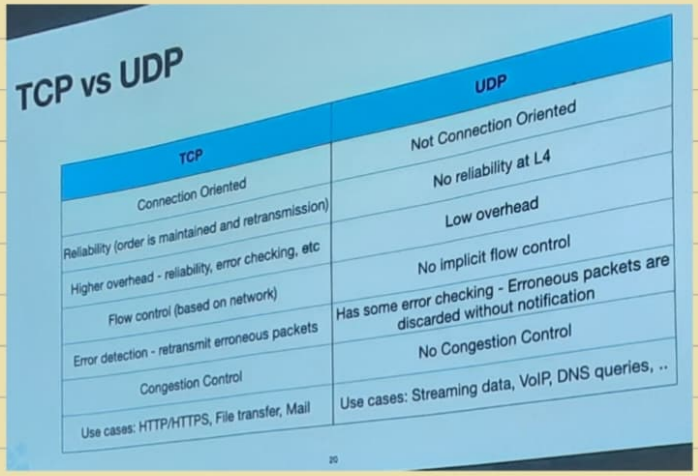
Process id	Port

← Mapping

- Socket → Abstraction → Allows communication
  ↳ API

Stream → continuous (TCP)
↳ Mutual connection (strict) + Connection oriented

Datagram → No memory + No persistent connection (Stateless connection)
↳ UDP
↳ Unrealiable (by default)



Flow control : Control speed of data transmission.

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- UDP ⟶ Unreliable at L4 ∵ Fire and forget, i.e. responsible for giving data but not ensuring recieving.

Headers differentiate TCP and UDP.
↳ 32 bits ↳ lightweight (8 bytes)

TCP ⟶ 3-way Handshake (Agreement)
L4 ensures packets getting delivered in-order.
↳ Transmission Rate

- TCP Header :
  ① Sequence number : Allows sequencing of data
  ② Acknowledgement number : Sent back from reciever, to let the sender know that data was recieved and is waiting for next data.

Data is sent ⟶ Wait to recieve acknowledgement (OS starts a timer)

↙ Data is recieved
↘ Data is lost ⟶ Retransmit the data

OS does not want to waste resources
⇒ ∴ Starts a timer and if it elapses ⇒ Timed Out

Data can be lost even when acknowledgement no. > sequence no.

Acknowledgement cache ⟶ Prevents Deadlock
↳ Sender sends & waits for response (ACK)
Reciever sends back & still on the way

"IIIT Timer ⟶ 1½ hour delay"

- RTT : Round Trip Time
  Sample RTT : How much time
  Estimated RTT : Keep updating estimation on every sample
  $(1-\alpha)$ ERRT + (α) SRTT
  ↳ If very high, ∴ Works only on current & not past
  Default : 0.125

DeviationRTT : Worst case scenario, can happen but rarely

$$DRTT = (1-\beta)\,DRTT + \beta \times |SRTT - ERTT|$$

↳ Weighted moving range ($\beta : 0.75$)

$$\text{Timeout Interval} = ERTT + (4 \times DRTT)$$

- 3 → 85 % confidence
- 2 → 65 % confidence
- 4 → 99 % confidence

[Based on experiments]

② Check acknowledgement constantly for every packet ?

Disadvantage (if check) ⟹ High overhead & Latency

Solution ⟹ Delayed Acknowledgement

↳ Wait after sending multiple packets and reciever sends ACK

To prevent deadlocks :

Timer on the reciever end

③ Window : Flow Control

Reciever has a limit to recieve data & then waiting time.

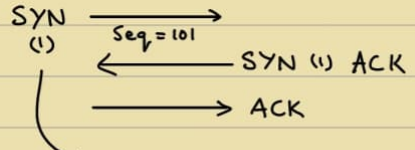
Send a 0 byte data to know whether reciever can recieve data.

Reciever & Sender ⟶ Both send data, but have seperate values of ACK.

Sequence number ⟶ Any random number within the bounds.

↳ Decided after agreement

3-way handshake ⟶ " Can we start connecting ? "



1 : Setting the SYN bit, i.e. sending the acknowledgement

Agreement on Seq. no. from both sides

ACK → 1 : Accepts the mutual seq. no.

SYN ACK : Mutual Agreement & that both recieve data

SYN : 1 ⟹ Connection establishes and sender initiates sending

Closing connection : FIN and RST flags

Mutual way termination (4-way exchange)

Forceful way termination

No acknowledgement ∴ connection is terminated

Note : Size of TCP Header : 30 bits to 60 bits (Option Speed)

P bit : [Push bit]

Send data ⟹ Reciever gets and puts in buffer (Generally)

P = 1 ⇒ Data is directly sent to process , No buffer used

U bit : [Urgent bit]

Some packets sent need to be processed urgently and rest can be processed later

- Memory :

Every Process requires memory.

Memory Virtualization ⟶ Every process feels like its getting entire memory.

Some of the RAM is used up by OS , code and data . Rest of the memory is usable RAM , i.e. our code and data.

Multiple Processes running at the same time ⟹ fast

Ways to achieve process virtualisation :

(1) Partitioning OS into slots

Disadvantages : - Other slots data can be accessed.

- Each process may not be fixed amount of data.

{ Saturday : 8:30 Tutorial ⟶ Class }

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Flow control : Just b/w sender and reciever (L4)

Congestion control : C-bit , In the network layer

- Memory virtualization :

  e.g. Supermarket vs e-shopping

  Every process ⟵ Address space
  ⟶ Virtual Address Space (VA) , mapped to Physical memory

  *p = 3 ⟹ &p ⟵ Virtual memory

  Note : Only 1 Physical Memory ⟵ Physical Address (PA)

  Processes might not always stay in Physical memory , it can be in disk as well from overflowing . (Note : Swap Space)

  * Translation : Memory Management Unit (MMU)
    ⟶ Fast
    ⟶ Conversion + Fetch

- Goals of Virtualization :
  (1) Transparancy ⟵ Each process gets an illusion of infinite space
  (2) Efficiency ⟵ Use hardware
  (3) Protection

- For Process Virtualization :
  Mechanism : Limited Directed Execution (LDE)
  Policies : Scheduling Algorithms

- For Memory Virtualization :
  Assumptions :
  (1) User address space is contiguous in Physical Memory.
  Process ⟶ Block
  (2) Size of Address space is too big , less than size of Physical memory
  Process ⟶ No overflow
  (3) Every Address space is of equal size.

Memory Virtualisation :

Q) How do I divide Physical Address space.?

Q) How do I map Virtual Address to Physical Address ?

Reference the variable ← Stack

Note : Base and Bound approach :

→ Values are stored in MMU

Then, Add Base Address

VA → Process starts at 0

But PA : Process = 0 + Base value

= VA + Base

Accessing after bound → Fault : Address out-of-bound error

Allocation : Dynamic ⟹ Dynamic Relocation

VA to PA : Address Translation

→ MMU not OS

* MMU → @ Context-Switch

→ Has only 1 pair of Base and Bound

* Segmentation : (Memory Management System)

Code, Stack & Heap - Generalised Base and Bounds

∀ Segment : Has a base and bound, Not Fixed

→ Dividing PA space

Segment Registers in MMU.

First → Identify which segment

* Address Translation :

Stack → goes up ∴ Subtract

Real Address (PA) = VA + base address

Code : No offset

Heap : Offset

Calculation

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VA → Every Process

Map : VA → PA : Base and Bounds → Disadvantages :

VA → 0 to Max

(1) Fragmentation

(2) Unused Memory

- Identify segment :

(1) Implicit :

Stack

(2) Explicit : (VA → 14 bits)

First 2 bits → Segment (12-13)

Rest → Offset (0-11)

00 : Code

01 : Heap

11 : Stack

∵ Bits are unused ⇒ Code and Heap have same bit in some OS.

- Simple Address Translation :

Many OS : Segmentation ✗

Paging ✓

PA : Base + Offset

Offset → Data − Max stack data

-

Segmentation

Fine-grained
- Small size segments
- High overhead

Course-grained
- Large size segments

When different sized segments, Disadvantage :

External Fragmentation ← Extra memory is available

But not contiguous, i.e. has holes

Defragmentation : High overhead

Note : Free space management algorithm, Closest Fit, First fit are algorithms

Fixed size segments ⟹ Code, Heap & Stack ×

↳ Disadvantage : Internal Fragmentation

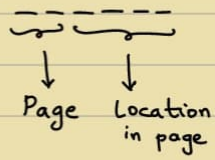
↳ Better than external fragmentation

Page : Fixed size segment in a process

Paging : VA and PA are both divided

∴ If 4 pages → $2^{②}$: 2 bits

64 bytes VA → $2^{⑥}$: 6 bits



VPN : Virtual Page Number

PFN : Page Frame Number

} ← Every page has a number

In PA : Page Frame

In VA : Page

Advantage :

(1) Flexibility : OS finds a free page

(2) Simplicity : ∀ Process , ∃ Page table

↳ Mapping b/w VA and PA

Array can be used , e.g. VPN[i] = PFN_j

Note : Multiple VPNs can be mapped to a PFN (not same process)

If $Mem_{Process} > Mem_{RAM}$ ⟹ Swap in & Swap out from Hard disk

- Process of Translation :

VPN Offset

VA_5 VA_4 VA_3 VA_2 VA_1 VA_0

e.g. mov 21 %eax

$(21)_{10} = (10101)_2 = (010101)_2$

01 : VA space
0101 : offset } → & if $VPN_1 = PFN_7$ (from page table)

Then $\underbrace{010101}_{VA} \Rightarrow \underbrace{111\,0101}_{PA}$

$(1110101)_2 = (117)_{10}$

Page Table Base Register ⟶ store base of the address of the page table.

e.g. 32 bit address space with 4 KB pages
no. of bits for offset ?

$4\ KB = 2^{12} \Rightarrow$ 12 bits to represent

$32 - 12 = 20$ bits for mapping, i.e. VPN

$\therefore 2^{20}$ mappings per process

Each mapping ⟶ 4 bytes

$\therefore 2^{20} \times 4 = 4$ MB per process per page

If 100 processes $\Rightarrow$ 400 MB for address translation

→ Page Table

$\therefore$ Use pages to store page tables

- Page Table :

Page Table Entry (PTE)

VPN , PFN , V , P , Present , Dirty , Ref

- V → Valid Bit
- P → Protection Bit
- Present → If page in Physical memory or secondary
- Dirty → If page is modified (from after memory)
- Ref → If page is recently used

$\therefore$ Disadvantage : Fetch twice for every translation
(Extract memory)

- Efficient Translation : Caching

Translation lookaside Buffer (TLB) – Register

→ Address Translation cache

→ In MMU

PTBR
↓
Page Table
↓
PTE
↓
Find

9/9

TLB → Hit rate of cache ↑

Cache : VPN ⟷ PFN

Size of cache ↑ ⇒ Hit rate ↑

Size of page ↑ ⇒ Hit rate ↑

- Spatial locality : Space dimension , i.e. same space accessed
- Temporal locality : Time dimension , i.e. recently accessed.

- Hardware TLB handling - Hardware goes through the page in O(1).
  Software TLB handling - Hardware raises exception (TRAP)
  Context Switch from Hardware to Software

@ TLB miss handlers ⟶ stored in OS Kernel , avoiding translation (Physical memory)

TLB → Fully Associative cache
(No direct mapping , random)

- TLB :
  - Valid bit :

    Translation is valid or not → Page Table Entry case
    Process A to B ⇒ B should not use A's mapping ⟶ TLB case
    ∴ A → invalid

  - ASID :

    Every process has a Address Space Identifier .
    Mapping : Process id ⟷ ASID
    ASID → Small (∵ Cache)
    Process id → Disadvantage : Large amount of bits (32 to 64)

- (i) Segmentation + Paging } ⟶ Reduce Page Table size
  (ii) Encoding

Page size ↑ ⇒ Mapping ↓ but Internal Fragmentation ,
lot of empty sizes will be left

(i) Segmentation + Paging ⟶ Getting best of both ?

Segmentation → External Fragmentation
Paging → Internal Fragmentation
} Issues

Page Table → stored in RAM
↳ divided into pages
↳ Stored using Pages

Page Table contains mappings, entries.
Each mapping is not important, only valid entries are
Invalid entries → don't store, store only when it becomes valid.
Valid entries → store
∴ Multi - Page Table (Tree-like)
i.e. Meta-index, multiple pointers heirarchy.

* [Explore 2-level Page Table
Study 3-Level Page Table] *

Page Directory → Data structure for multi-level page tables.

PDBR → less space (∵ Only valid PTEs are stored)

Disadvantage : Time overhead

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- Inverted Page Table : 1 page table per process
disadvantage : Linear scan is expensive

- OS creates Memory Heirarchy.
Physical Memory → Faster access
Less capacity
Disk → Slower access
More capacity

On-demand Pages
- Swap Space :
↳ Allocated in disk
↳ divided into blocks ≈ Page size

- Present bit :
1 : page is in the page table
0 : page is in the swap space

- Valid bit = 0 : Page is not in Physical memory ⇒ In disk

AKA Page Fault → Handler

Page Fault → Process moved to block state

Handler → What to swap, when & from where
AKA Page Replacement

- AMAT
- Optimal Replacement Policy (Theoretical Optimal)
  Belady Replacement Policy (But Not Practical)
  ↳ Possible future access

Reduce any policy to Belady for comparison

(i) FIFO → First in First Out

Cache size ↑ ⇒ no. of hits ↑ (Belady's Anamoly)
↳ Not always, depends on ordering of stream of data.

(ii) LRU → Least Recently used

- Valid bit = 1
- Access bit = 1
  ↓ Recently modified
  Access bit = 0 ← Evict a page

- Thrashing → Excessive Page fault Handlers

Sat → Tut (IMP) [Bonus Quiz]
Tues → Tut (Attendance) [MP2]
Fri → None

26/6

SMTP uses TCP

(Spotify – Web series)

Install Apache / ngnix webserver

→ Peer 2 Peer Network :

Spotify : Midserver + Peer2Peer

(Client server)

→ Application level Protocol :

HTTP : 80

HTTPS : 443 (HTTP over secure network)

∀ connection → 3 way Handshake (∵ HTTP runs on TCP)

↳ Stateless

- 1.0 HTTP ← For fetching every connection, connection is established.
  1.1 HTTP ← Maintain connection to fetch all objects and then terminates.
  2.0 HTTP & 3.0 HTTP

→ Cookies :

∵ HTTP is stateless, Data stored in browser cache.
Metadata is stored in server.

Web cache :

Browser cache
ISP cache
Regional cache
} If recently changed, use cache
Main server

- Traditionally,
  Client ⟷ Server : Takes time
  (Delay)

∴ Add cache at server.

Content distribution Networks (CDN) :

↳ Servers distributed among diff. locations.

(Logics : Potential viewership + location)

Enter Deep : Build smaller clusters in more sites

Bring Home : Build large cluster in lesser sites

Cache does not contain originals

↳ Conditional GET : Data from server would only be header, no body

HTTP 2.0 → Current

HTTP 3.0 → Future, but has support now.

→ Domain Name System (DNS) :

Translation b/w name and I/P address. Allows aliasing.

Uses UDP.

- UDP call to get I/P address & HTTP call happens, i.e. then client uses HTTP to send request to server.
- DNS record
- BIND → Turns machine to DNS server → Port 53
  i.e. machine acts like a directory, returns IP address
- Any request (HTTP, SMTP, etc) sent translates the hostname to IP address using DNS

- 1. Root DNS → Report to IANA
  ↳ 12 different Organizations are responsible for DNS creation, etc
  2. Top-level Domain (TLD) : .com, .org, .ai domains
  3. Authoritative Domain

- How it works : (Recursive Process), When you send a request to a host,
  1. Request is sent through Port 53
  2. Local DNS (ISP Provides) ← Cache kind-of
     If mapping exists, fetch. Else goes to next level (Recursion)
  3. Root DNS (if it has, it will give back, else send to next level)
  4. TLD DNS (if it has, it will give back, else send to next level)
  5. Authoritative DNS (Then, get I/P address and return to client)
  6. Then, you send a HTTP request to the server.

- DNS servers stores Resource Records (RR) : (name, value, type, ttl)
  ↳ tuple
  ↳ Each DNS record

ttl : time after which it needs to be updated from the root DNS.
↳ Time To Live

(i) A type DNS record : (Authoritative)

Name : Hostname

Value : I/P address

(AAA → IPV6
  A → IPV4)

(ii) CNAME type → Mapping one domain to another domain

↳ Canonical name record (alias)

(iii) NS → Named Server

Main server itself (Authoritative)

(iv) MX type → Mail

(mail.outlook.com)

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- Concurrency :

Multiple execution points sharing the same memory space

Same process – Multiple threads sharing same memory space.

- The threads synchronize with each other.
- Should not overlap ideally.

Thread : Smallest unit of execution.

Seperate P.C

Seperate stack of local variables

Creates child processes

Same address space / memory

Concurrency in Single core machine → Context switching

fork → child is created , PID diff. from that of parent.

Memory copy is created , code is shared.

exec → Child is created with same PID.

Replace the memory , ∴ diff. memory

Inter-process communication → reading / writing from shared files

Multiprocessing : Multiple processes executing in diff. cores of the CPU in the same time on the same data.

Scheduling → Thread level

Not Process level

- Kernel Space → Kernel level threads → Scheduled by Kernel

  User level threads → Scheduled by user , i.e. user level libraries ← pthreads

- Concurrency : Interleaving
  - Context - switching in case of single core
  - Dealing with lot of things at once.

  Parallelism : At same point of time → Multiple execution

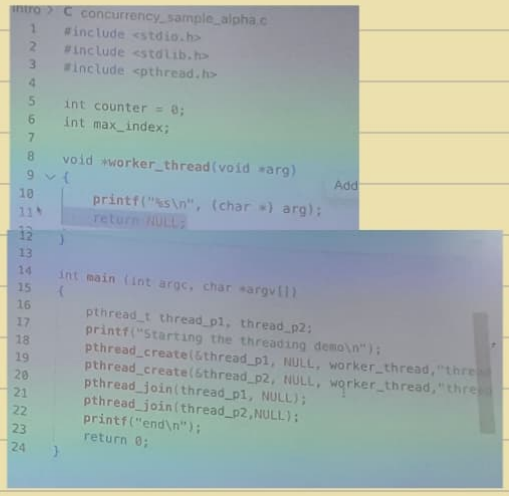
  ∴ Multiple cores are required.
  - Doing lot of things at once
  - Subset of concurrency

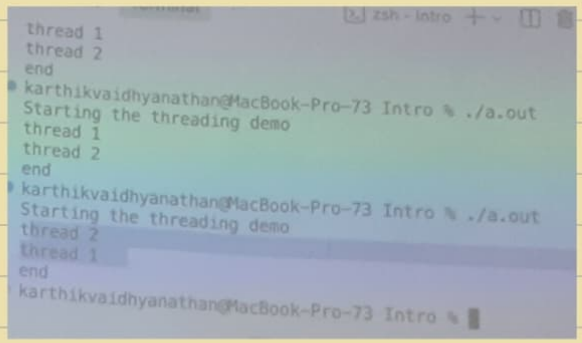
- TCB : For every thread, ∃ TCB

  Kernel level threads : Scheduled by OS , Handles system calls.

  User threads : What process a developer writes , e.g. processing a file.

- Start routine





↳ Non - deterministic + No wait()

Depends on scheduler

Reason : Race condition

↳ access b/w shared variables

No synchronisation

Thread scheduling → Process level is done by user

↳ Kernel level is done by OS

What if there is a shared variable ?

30/9

Elective course → Compilers

→ If shared variable is used,

Interrupt → Context switching b/w threads

∵ Multiple threads are trying to get access to shared memory.

Race condition : Multiple threads are trying to compete against each other, but we don't know which thread gets access first.

Critical Section : That piece of code/memory which is shared b/w threads

- Concurrency bugs can even cost lives.

- Disabling interrupts → Not a good option (∵ Kernel has no access)

Mutual exclusion → Execute them atomically

↳ Hardware + Software

↳ Every thread should have access the critical section

↳ Build support for synchronisation

∴ Locking can be used ← flag variable

Datatype : Struct ← Software Lock

convention : mutex (mutual exclusion)

Lock → Execute (Increment) → Release lock [User Process]

Lock → 1
Unlock → 0
} Owner calls

- Building a lock :

```
if (flag == 0)
  flag = 1
```
} → Two threads can check at the same time.
∴ Not a good idea

Better option :

```
while (flag == 1) {....}
flag = 0
```

* Lock should support mutual exclusion. (Mutual Exclusion)
* Lock should allow every thread to use critical section. (Fairness)
* Locking should be highly efficient. (Efficiency)

- If flag = 1 ⇒ Waiting
  flag = 0 ⇒ Use the critical section & set flag = 1 so that other threads can't use.

- Problem :
  (1) Two instructions - Fetch and Compare - No atomicity
  (2) If one thread comes out of while loop but at the same time, another thread changes flag variable, making it seem like both threads have access.
  Sol : 1 flag for every thread in the struct
  (But what if there are 100s of threads ?)
  (3) Spin-based lock : When both threads have flag set to 1.
  (i.e. when context switching occurs just after flag check)

Sol :
(1) Test and Set :
  ↳ Written in Assembly ∴ Atomic
  ↳ Testing of old values while setting the main to a new value.

∴ while (flag == 1) ×
  while (TestAndSet(&lock → flag, 1)) ✓
    ↳ ∴ Only one thread gets access at a time

Problem : There can still be starvation.
  Every thread can get into blocked state

(2) Compare And Swap :
  Test whether the actual value is equal to as expected.

```
while ( TestAndSet(&lock → flag, 1) == 1)
while ( CompareAndSwap(&lock → flag, 0, 1) == 1)
```

e.g. Linked list ⟶ {1, 2, 3}
  Multiple threads can operate on a particular node, for ex, list[0]
  While updating, then lock check occurs.
  In case of TestAndSet, list[0] is blocked by every other thread.

(3) Load-linked and Store-conditional :
  - Used by ARM architechture
  - No other thread should have updated since the lock was acquired.
  - To overcome ABA problem (A → B → A, Seem like it was always A)

- Fairness:

Every thread should be able to access the shared data.
(Bakery's Algorithm, etc)
Intuition → Every thread has a token (counter++ and turned value returned)
∴ To know whose turn is next.

e.g. Fetch And Add → Ticket lock
(Peterson's Algorithm)

3/10

- Fetch and Add:
  - Turns and Tickets
    - Turns → Assign a value, Keep incrementing automatically
    - Tickets → Who is currently going to access
  - Initialise both to 0.
  - Returns old
  - Spin lock: CPU gets wasted (Not it's turn, so continuously waiting)
    - → Could be infinite, keep spinning forever
    - → Thread is always in ready state → Fast, Efficient, reliable
  - In Kernels → Spin lock

- Yield: Telling the CPU to take the access back.
  ∴ If there are 100 threads, each thread will go through atleast 1 yield cycle.
  But, There could be infinite yield.

- Park → Immediately put a thread to sleep
  Unpark → Use PID to wake a thread up. } Solaris

- Locking is fine. (Whenever ∃ Shared variables)
  But there should be synchronization b/w threads.
  pthread_join: Parent waits until child is finished.
  - Parent → Keeps spinning
  - pthread_join → Use shared variables

  But if there are more threads → Inefficient
  ∴ Condition variable → Queue
  Waiting in the queue until condition achieved
  (i) wait - Puts itself to sleep (Wait until condition is true)
  (ii) Signal - communicate that condition is achieved, ∴ Wake threads up.

Note : Whenever there are shared variables ⇒ use locks
Whenever there are multiple threads ⇒ use signal

Note : Lock is released just after parent puts itself into the queue, but just before sleep.
Lock is reacquired just after getting out of Condition queue.

- Curious wake-up :
  Child can randomly wake up even when condition is not fulfilled.
  ∴ Use while() instead of if()

- Producer-consumer problem : (Bounded Buffer Problem)
  - Producer : Produce to the common buffer
    Consumer : Take from the common buffer
  - e.g. Video streaming : Platform sends video (set of images) to user
    User recieves video (set of images) based on bandwidth
  - If no synchronisation → Race condition
  - Bounded Buffer → Shared resource
    (Synchronization to be developed)

(Other problems :
Reader-Writers problems
Dining Problem
Rotatory Problem)

Note : Consumer must not consume if buffer is empty
Producer must not produce if buffer is full.
∴ Send Signal to consumer everytime producer produces something.
Send Signal to producer everytime consumer consumes something.
& Add locking mechanism for done variable.

- 2 shared variables
  1 condition variable → count
  } → Works only for 1 producer - 1 consumer

Which consumer to wake up ?
Problem : Everyone goes to sleep
∴ Use 2 condition variables → fill and empty.
↳ Producer waits on empty condition
↳ Consumer waits on fill condition
(OR) vice-versa

∴ Switch the logic instead of using 1 conditional variable, SOLVED!
i.e. Producer - Consumer problem is solved using Fill and Empty.
(Everyone going to sleep issue)

BUT now we have 2 primitives, i.e. lock and conditional variable.
Can be used to solve any conferencing problem.
Dijkstra – "Why do we need both?"

→ Make something that provides support for both
i.e. Semaphore : Structure that provides both forms

- REVISE before next class (Book)

Tue → No class
∴ Next class → Fri